

# SUHAAS CHOUDARY MALLAVARAPU

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## SUMMARY

Computer Science sophomore with hands-on experience building ML-integrated applications and scalable backend systems and data architectures using Python, TensorFlow/Keras, and MongoDB. Solid foundation in data structures, OOP, and software engineering fundamentals. Seeking a software engineering internship to contribute to backend systems, AI/ML integrations, and scalable application development.

## EDUCATION

**B.Tech in Computer Science and Engineering — SRM University AP, Andhra Pradesh** Aug 2024 – Present

CGPA: 7.4 | Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks

Class XII – MPC, Bhashyam Junior College | 86%

2024

Class X – ICSE, Loyola Public School | 90%

2022

## TECHNICAL SKILLS

**Languages:** Python, C, C++

**AI / ML:** TensorFlow/Keras, NumPy, OpenCV

**Backend & Databases:** MongoDB, PyMongo, modular backend development, CRUD operations

**Tools & Platforms:** Git, GitHub, Streamlit, VS Code, CLI

**Core CS:** Data Structures, Algorithms, Object-Oriented Programming, Software Engineering Fundamentals, Debugging, Problem Solving

## PROJECTS

### Handwritten Digit Recognition (CNN)

Python, TensorFlow/Keras, Streamlit, MNIST

GitHub: [github.com/suhaas1126/CodeAlpha\\_HandwrittenRecognition](https://github.com/suhaas1126/CodeAlpha_HandwrittenRecognition)

- Trained a CNN on the MNIST dataset (~60,000 samples) achieving 98%+ validation accuracy; applied Conv2D, MaxPooling, and Dropout layers to improve generalization.
- Built an interactive Streamlit web app with a real-time drawing canvas where users sketch digits and receive instant predictions alongside per-class probability visualization.
- Designed a preprocessing pipeline to normalize, reshape, and invert drawn inputs, bridging the gap between freehand sketches and the MNIST training distribution.
- Structured project into distinct modules for model training, inference, and UI rendering — enabling independent testing of each layer.

### Student Record Management System

Python, C, MongoDB

GitHub: [github.com/suhaas1126/Student-record-management-system](https://github.com/suhaas1126/Student-record-management-system)

- Built a backend system with full CRUD functionality using a layered architecture that separates data access, business logic, and the interface layer for cleaner code organization.
- Improved data retrieval performance across 1,000+ records by replacing linear searches with dictionary-based lookups and sorted lists, noticeably reducing query response time.
- Designed optimized schemas and queries using PyMongo to ensure efficient read/write operations and logical data organization as record volume grows.
- Kept modules independently testable by following single-responsibility principles, making it straightforward to add new features without breaking existing functionality.

## CERTIFICATIONS & ACTIVITIES

- MongoDB Associate Developer Certification (2024)** — validated proficiency in NoSQL data modeling, aggregation pipelines, indexing, and CRUD operations across 5 assessment domains.
- Science & Technology YouTube Channel** — produced 50+ educational videos on emerging tech topics, demonstrating entrepreneurial passion, self-directed learning, and excellent technical communication skills.